

yolov10n Hyperparameter Tuning Report

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1. Optimization Overview

Property	Value
Model	yolov10n
Dataset	bdd100k_yolo_limited
Total Trials	12
Completed Trials	12
Failed Trials	0
Best Trial	0
Best mAP@0.5	0.1530
Optimization Duration	0:21:14.985095

2. Optimization Configuration

Parameter	Value
Total Trials	12
Epochs per Trial	6
Batch Size	64
Image Size	640
Startup Trials (TPE)	5
Device	cuda
Number of Classes	10
Train Images	945
Val Images	611

3. Best Hyperparameters

Parameter	Value	Description
optimizer	Adam	Optimization algorithm
lr0	0.001040	Initial learning rate
lrf	0.162407	Final learning rate factor
momentum	0.868719	SGD momentum / Adam beta1
weight_decay	0.000013	Weight decay (L2 penalty)
warmup_epochs	3	Warmup epochs
warmup_momentum	0.770502	Warmup momentum
warmup_bias_lr	0.070807	
hsv_h	0.002058	HSV-Hue augmentation
hsv_s	0.787964	HSV-Saturation augmentation
hsv_v	0.732977	HSV-Value augmentation
scale	0.827403	Scale augmentation
translate	0.036365	Translation augmentation
mosaic	0.591702	Mosaic augmentation
mixup	0.060848	Mixup augmentation
box	7.148539	Box loss gain
cls	1.147918	Classification loss gain
dfl	0.936844	Distribution focal loss gain

4. Top 20 Trials Performance

#	mAP@0.5	Optimizer	lr0	momentum	mixup	mosaic
1	0.1530	Adam	0.0010	0.869	0.06	0.59
2	0.1225	Adam	0.0019	0.888	0.11	0.60
3	0.1088	Adam	0.0001	0.909	0.03	0.70
4	0.1023	Adam	0.0019	0.882	0.16	0.87
5	0.0977	Adam	0.0030	0.913	0.06	0.54
6	0.0970	Adam	0.0006	0.898	0.04	0.69
7	0.0428	SGD	0.0010	0.871	0.08	0.59
8	0.0010	SGD	0.0004	0.944	0.06	0.90
9	0.0010	AdamW	0.0004	0.915	0.15	0.86
10	0.0010	Adam	0.0035	0.930	0.02	0.96
11	0.0010	SGD	0.0001	0.889	0.10	0.75
12	0.0010	Adam	0.0028	0.878	0.06	0.83

4.1 Detailed Hyperparameters - Top 5 Trials

Rank 1: Trial 0 (mAP@0.5: 0.1530)

box=7.148539, cls=1.147918, dfl=0.936844, hsv_h=0.002058, hsv_s=0.787964, hsv_v=0.732977, lr0=0.001040, lrf=0.162407, mixup=0.060848, momentum=0.868719, mosaic=0.591702, optimizer=Adam, scale=0.827403, translate=0.036365, warmup_bias_lr=0.070807, warmup_epochs=3, warmup_momentum=0.770502, weight_decay=0.000013

Rank 2: Trial 7 (mAP@0.5: 0.1225)

box=6.613713, cls=1.111203, dfl=1.904247, hsv_h=0.035141, hsv_s=0.731873, hsv_v=0.770032, lr0=0.001917, lrf=0.215342, mixup=0.113125, momentum=0.887982, mosaic=0.599302, optimizer=Adam, scale=0.808287, translate=0.058469, warmup_bias_lr=0.028088, warmup_epochs=3, warmup_momentum=0.655529, weight_decay=0.000023

Rank 3: Trial 9 (mAP@0.5: 0.1088)

box=8.150549, cls=1.024536, dfl=1.173397, hsv_h=0.020077, hsv_s=0.497664, hsv_v=0.740270, lr0=0.000129, lrf=0.222123, mixup=0.033707, momentum=0.908993, mosaic=0.704247, optimizer=Adam, scale=0.776039, translate=0.042970, warmup_bias_lr=0.088995, warmup_epochs=3, warmup_momentum=0.809426, weight_decay=0.000020

Rank 4: Trial 11 (mAP@0.5: 0.1023)

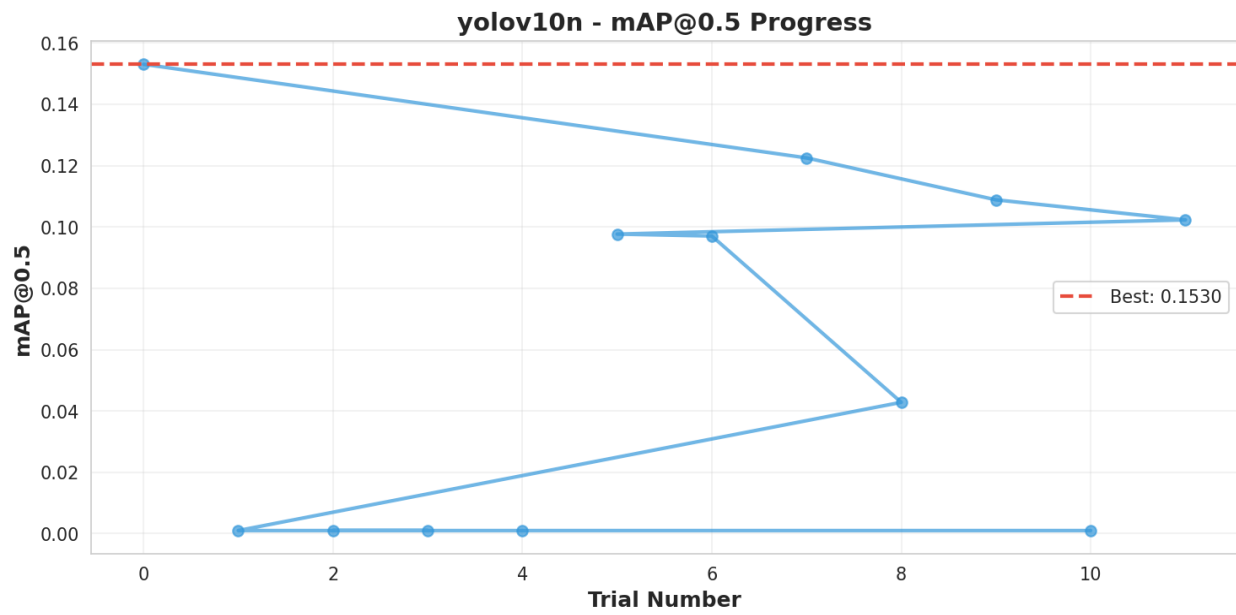
box=7.112519, cls=0.877320, dfl=1.607794, hsv_h=0.029006, hsv_s=0.730006, hsv_v=0.776093, lr0=0.001934, lrf=0.278688, mixup=0.160591, momentum=0.881797, mosaic=0.867038, optimizer=Adam, scale=0.732008, translate=0.087208, warmup_bias_lr=0.038575, warmup_epochs=3, warmup_momentum=0.534897, weight_decay=0.000020

Rank 5: Trial 5 (mAP@0.5: 0.0977)

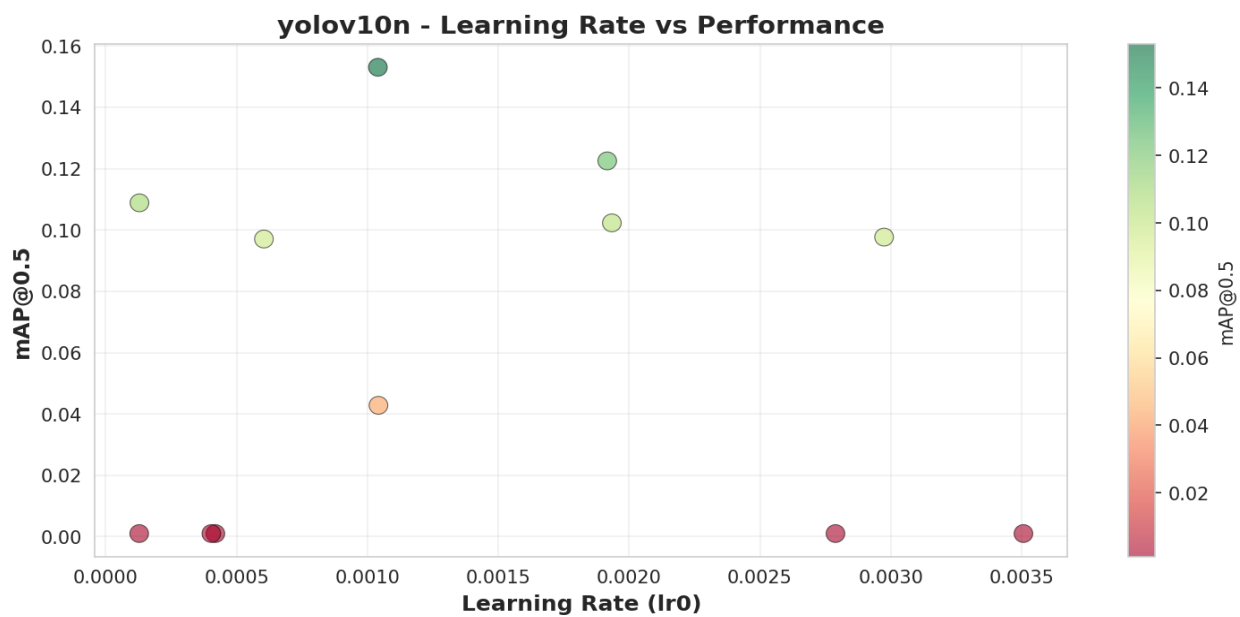
box=7.527204, cls=1.027950, dfl=0.861551, hsv_h=0.045387, hsv_s=0.783606, hsv_v=0.715865, lr0=0.002974, lrf=0.122744, mixup=0.064797, momentum=0.912982, mosaic=0.540900, optimizer=Adam, scale=0.764245, translate=0.021880, warmup_bias_lr=0.096397, warmup_epochs=3, warmup_momentum=0.797665, weight_decay=0.000057

5. Optimization Visualizations

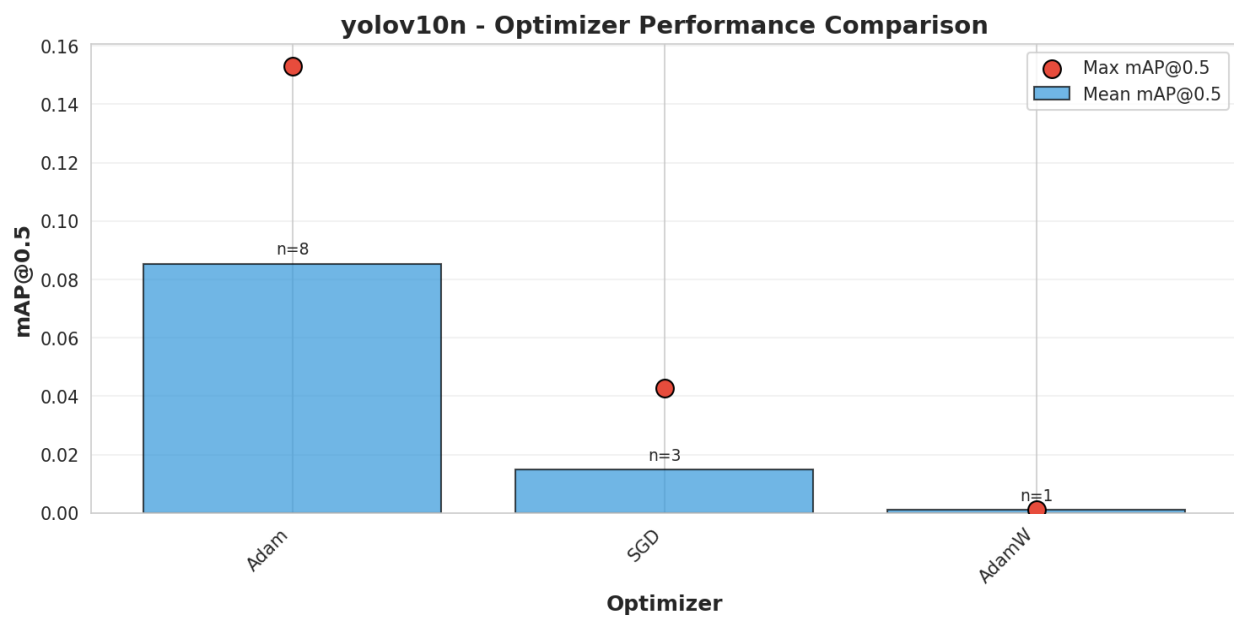
5.1 mAP@0.5 Progress Over Trials



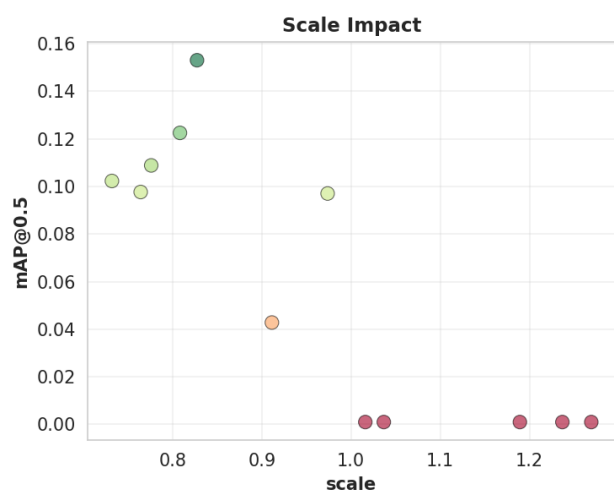
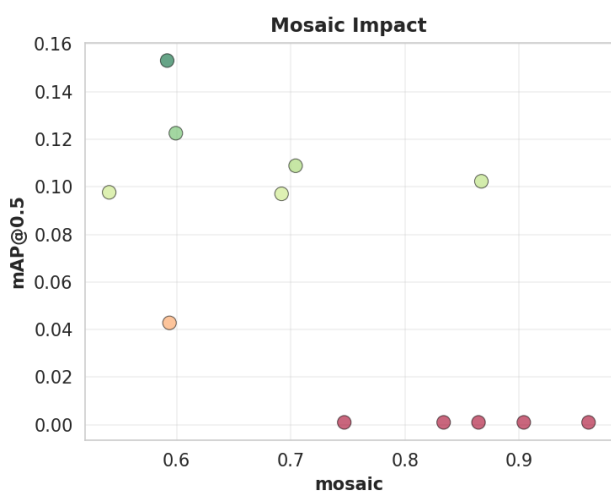
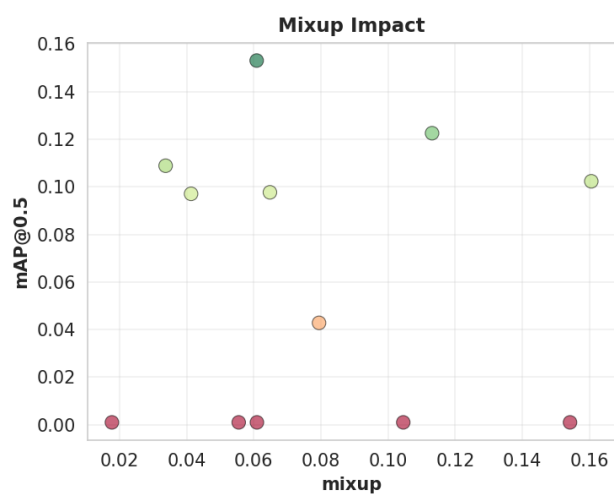
5.2 Learning Rate Impact on Performance



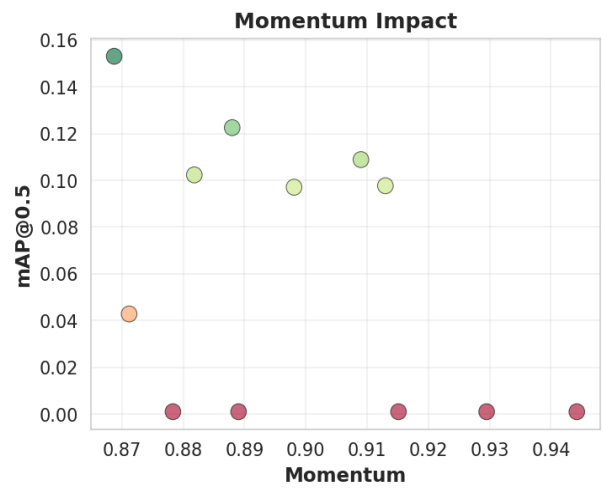
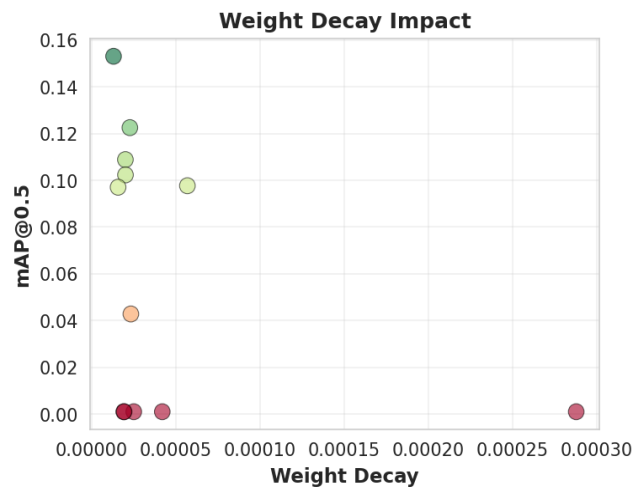
5.3 Optimizer Performance Comparison



5.4 Augmentation Parameters Impact



5.5 Regularization Parameters Impact



6. All Trials Summary

Metric	Value
Completed Trials	12
Best mAP@0.5	0.1530
Worst mAP@0.5	0.0010
Mean mAP@0.5	0.0608
Std Dev mAP@0.5	0.0582
Median mAP@0.5	0.0699